Learning how to prove Workshop Week 5

Permutations

Even and Odd Permutations

1. The **sign of a permutation** $\sigma \in S_n$ is defined as

$$\text{sgn}\ \sigma = \begin{cases} 1 & \text{if } \sigma \text{ is even} \\ -1 & \text{if } \sigma \text{ is odd} \end{cases}$$

Prove that $\text{sgn}\,(\sigma\tau) = \text{sgn}\ \sigma\,\text{sgn}\ \tau$.

Solution: Let us represent even permutations by E, while odd permutations are referred by O. We have the following cases displayed in the table below.

σ	τ	$\sigma\tau$	sgn σ	sgn τ	sgn $\sigma \cdot$ sgn τ	sgn$(\sigma\tau)$
E	E	E	1	1	1	1
E	O	O	1	-1	-1	-1
O	E	O	-1	1	-1	-1
O	O	E	-1	-1	1	1

In any case, it holds that

$$\text{sgn}\,(\sigma\tau) = \text{sgn}\ \sigma\,\text{sgn}\ \tau,$$

finishing the proof.

2. **Prove that every r-cycle is a product of $r-1$ transpositions:**

$$(k_1\ k_2 \ldots k_r) = (k_1\ k_2)(k_2\ k_3)\ldots(k_{r-1}\ k_r) = (k_1\ k_r)(k_1\ k_{r-1})\ldots(k_1\ k_2) \qquad (1)$$

Without factorising into transpositions, find the parity of $\sigma = (1\ 3\ 2\ 5\ 6)(2\ 3)(4\ 6\ 5\ 1\ 2)$.

Solution: Let $\sigma = (k_1\ k_2)(k_2\ k_3)\ldots(k_{r-1}\ k_r)$ and $\tau = (k_1\ k_r)(k_1\ k_{r-1})\ldots(k_1\ k_2)$. In order to prove (1), we will show that

$$\sigma(k) = \tau(k) = (k_1\ k_2 \ldots k_r)(k), \qquad (2)$$

for every $k \in X_n$. (Recall that a permutation is a bijective map $X_n \to X_n$). If $k \in X_n - \{k_1, \ldots, k_r\}$ then three permutation fix k, and (1) holds. The action of the cycle $(k_1\ k_2 \ldots k_r)$ on each k_i is clear. Let us check that it coincides with $\sigma(k_i)$ and $\tau(k_i)$:

$$\sigma : k_1 \mapsto k_1 \mapsto \ldots \mapsto k_1 \mapsto k_2,$$

$$\sigma : k_2 \mapsto k_2 \mapsto \ldots \mapsto k_2 \mapsto k_3 \mapsto k_3,$$
$$\vdots$$
$$\sigma : k_{r-1} \mapsto k_r \mapsto \ldots \mapsto k_r,$$
$$\sigma : k_r \mapsto k_{r-1} \mapsto k_{r-2} \mapsto \ldots \mapsto k_2 \mapsto k_1,$$

$$\tau : k_1 \mapsto k_2 \mapsto \ldots \mapsto k_2,$$
$$\tau : k_2 \mapsto k_1 \mapsto k_3 \mapsto \ldots \mapsto k_3,$$
$$\vdots$$
$$\tau : k_{r-1} \mapsto k_{r-1} \mapsto \ldots \mapsto k_{r-1} \mapsto k_1 \mapsto k_r.$$

It concludes the proof of (2).

Notice that σ is the product of three cycles (one which are not disjoint. This will not affect us when finding the parity since the parity of a permutation is an invariant (i.e. any two factorisations into transpositions will both have either an even number of transpositions of an odd number of transpositions; this is what the *Parity Theorem* says). Using the previous exercise we can write σ as a product of $4 + 1 + 4 = 9$ transpositions; so σ is odd by the *Parity Theorem*.